Definition 0.0.1. A map $f : \mathbb{R} \to \mathbb{R}$ is called *Monotonic* if:

$$\text{for any } a < b < c,\ f(b) \in [\min\{f(a), f(c)\}, \max\{f(a), f(c)\}]$$

This definition is equivalent to f is monotonically increasing or monotonically decreasing.
If monotonically increasing or decreasing, clearly monotonic.
Conversely, if f is not monotonically increasing and decreasing, then,

$$\text{there exists } x_1 < x_2 \text{ such that } f(x_1) > f(x_2)$$

which means that f is not monotone increasing, and

$$\text{there exists } y_1 < y_2 \text{ such that } f(y_1) < f(y_2)$$

which means that f is not monotone decreasing.
If $x_1 = y_1$ or $x_1 = y_2$ or $x_2 = y_1$ or $x_2 = y_2$, then directly f is not monotonic. Assume there are distinct.
Then, there are six cases:

1. $x_1 < x_2 < y_1 < y_2$:
   - $f(x_2) < f(y_2) \implies f(x_2) < \min\{f(x_1), f(y_2)\}$ on $x_1 < x_2 < y_2$
   - $f(x_2) \geq f(y_2) \implies f(y_1) < \min\{f(x_2), f(y_2)\}$ on $x_2 < y_1 < y_2$
2. $x_1 < y_1 < x_2 < y_2$:
   - $f(y_1) \geq f(x_1) \implies f(y_1) > \max\{f(x_1), f(x_2)\}$ on $x_1 < y_1 < x_2$
   - $f(y_1) < f(x_1) \implies f(y_1) < \min\{f(x_1), f(y_2)\}$ on $x_1 < y_1 < y_2$
3. $x_1 < y_1 < y_2 < x_2$:
   - $f(y_1) < f(x_1) \implies f(y_1) < \min\{f(x_1), f(y_2)\}$ on $x_1 < y_1 < y_2$
   - $f(y_1) \geq f(x_1) \implies f(y_2) > \max\{f(x_1), f(x_2)\}$ on $x_1 < y_2 < x_2$
4. $y_1 < y_2 < x_1 < x_2$:
   - $f(y_2) > f(x_2) \implies f(y_2) > \max\{f(y_1), f(x_2)\}$ on $y_1 < y_2 < x_2$
   - $f(y_2) \leq f(x_2) \implies f(x_1) > \max\{f(y_2), f(x_2)\}$ on $y_2 < x_1 < x_2$
5. $y_1 < x_1 < y_2 < x_2$:
   - $f(x_1) \leq f(y_1) \implies f(x_1) < \min\{f(y_1), f(y_2)\}$ on $y_1 < x_1 < y_2$
   - $f(x_1) > f(y_1) \wedge f(x_1) \leq f(y_2) \implies f(y_2) > \max\{f(x_1), f(x_2)\}$ on $x_1 < y_2 < x_2$
   - $f(x_1) > f(y_1) \wedge f(x_1) > f(y_2) \implies f(x_1) > \max\{f(y_1), f(y_2)\}$ on $y_1 < x_1 < y_2$
6. $y_1 < x_1 < x_2 < y_2$:
   - $f(x_1) > f(y_1) \implies f(x_1) > \max\{f(y_1), f(x_2)\}$ on $y_1 < x_1 < x_2$
   - $f(x_1) \leq f(y_1) \implies f(x_2) < \min\{f(y_1), f(y_2)\}$ on $y_1 < x_2 < y_2$

0.1 Open Continuous map is Injective

> **Theorem 0.1.1.** If $f : \mathbb{R} \to \mathbb{R}$ is Open Continuous map, then f is injective.

Proof. First, show that f is Monotonic. Suppose that f is <u>Not</u> monotonic.
Thus, there exists $x_1 < x_2 < x_3$ such that

$$f(x_2) \notin [\min\{f(x_1), f(x_3)\}, \max\{f(x_1), f(x_3)\}]$$

Therefore, there exists $t \in (x_1, x_3)$ such that

$$f(t) = \max_{x \in [x_1, x_3]} f(x) \text{ or } f(t) = \min_{x \in [x_1, x_3]} f(x)$$

by the Min-Max Theorem. Since the f is open map, $f[(x_1, x_3)]$ is open.
For $f(t) \in f[(x_1, x_3)]$, there exists $\delta > 0$ such that $(f(t) - \delta, f(t) + \delta) \subset f[(x_1, x_3)]$.
But, this is contradiction to $f(t)$ is maximum or minimum of $f[(x_1, x_3)]$. Consequently, f is monotonic.
Thus show injectivity, suppose that for some $a < b$, $f|_{(a,b)}$ is constant. But, then, $f[(a, b)]$ is singletone, closed.
It is contradiction to f is open map. Finally, f is strictly increasing or strictly decreasing, thus injection. $\square$